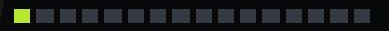


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Keep a useful repertoire.

When the useful AI output is a repertoire

Fixed
goal

Worked
exercise

Stated
challenge

Open
goal

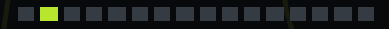
Explore
with prompts

Own
question

Guided

Self-directed

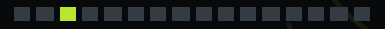
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The design question

A ranked list tells you which candidate won. A map can show which kinds of candidate are still worth having. Suppose you are designing a learning activity. A guided exercise and an open investigation can both be useful, but for different purposes. Scoring them against one preferred experience may eliminate the very difference a learner would later need.

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The design question

I would make that difference visible before choosing. Put guidance on one axis and destination openness on another. Then place a worked exercise, a stated challenge, exploration with prompts and an investigation driven by the learner's own question on the map.

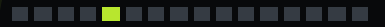
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The map is a design decision

This is a manual product exercise, not an evolutionary-computation experiment. It is useful because it forces the team to name differences in what a person can do. Four pieces of similar copy are not four different learning experiences.

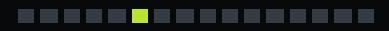
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The map is a design decision

The axes also carry assumptions. A map of setup effort and collaboration would preserve a different set of alternatives. Choosing a map is choosing which differences receive attention. No diagram can make that decision for the people building the product.

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The map is a design decision

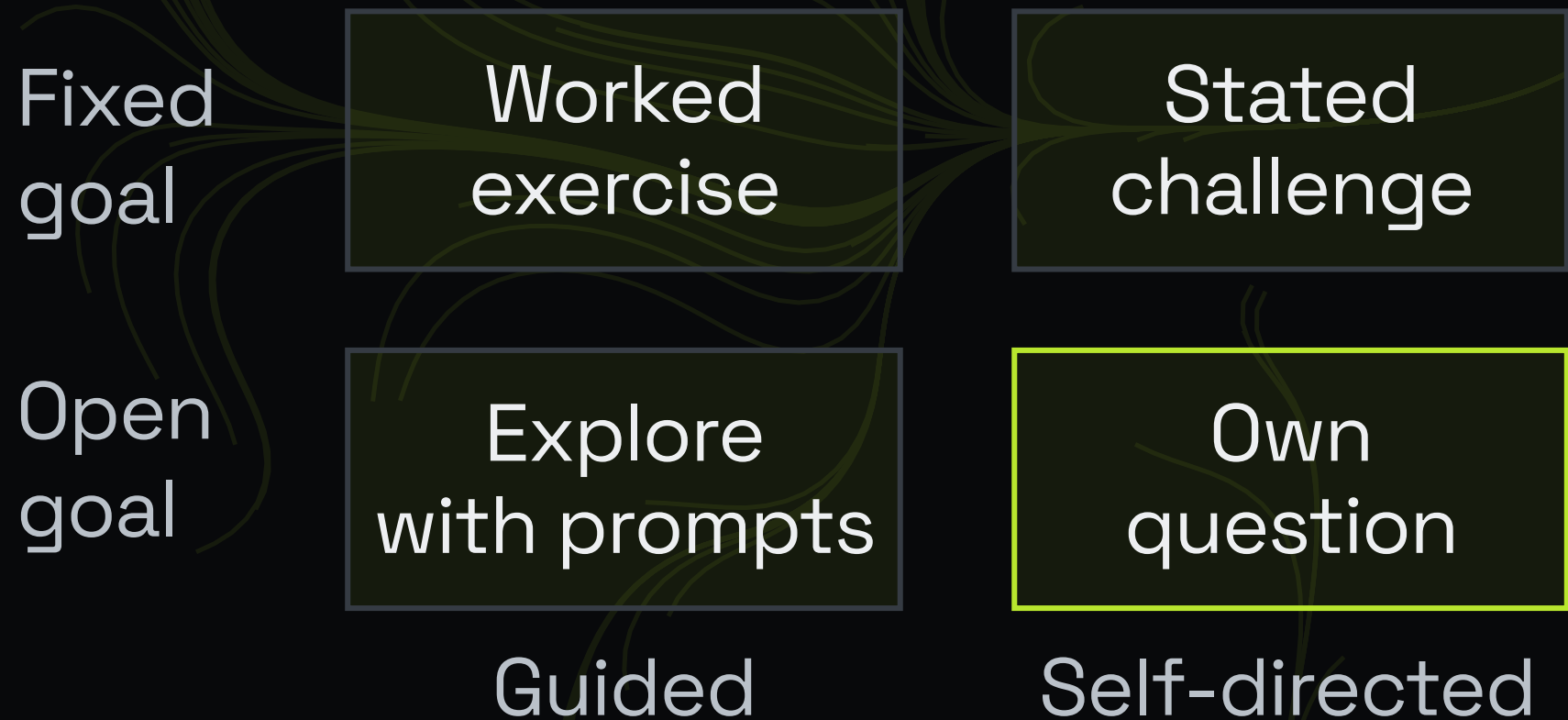
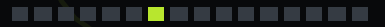


Figure 1. A proposed learning-activity map. Positions express design categories, not measured behavior or educational effectiveness.

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What the research mechanism adds

MAP-Elites offers a more precise computational idea: search for strong candidates across regions of a user-chosen feature space, retaining an archive rather than only an overall winner. In its basic procedure, a new candidate can fill an empty region or replace that region's incumbent when its performance is better. Variation of archived candidates continues the search. Mouret and Clune, 2015.

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What the research mechanism adds

Two things have different jobs here. A descriptor says where a candidate belongs; a quality measure judges its performance. Let x be a candidate, $b(x)$ its descriptor region and $f(x)$ its task-specific quality score. The archive entry $A[b(x)]$ holds the current candidate for that region. Replacement compares $f(\text{new})$ with $f(\text{current})$ within the same region. The score's units come from the chosen task; there is no universal unit of quality.

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What the research mechanism adds

The manual worksheet borrows the repertoire idea. It does not implement the algorithm, establish its search properties or inherit its experimental results. The foundational paper is also an early draft with specific experimental domains; it is not a present-day benchmark of agent products.

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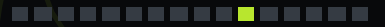
What the research mechanism adds



Other regions keep their candidates.

Figure 2. An illustrative archive update. A better candidate replaces an incumbent in the same region; other regions retain their alternatives. No scores or outcomes are measured here.

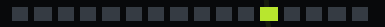
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Choose the output the decision needs

A ranked answer can be appropriate when the objective is settled and the person wants one choice. A diverse shortlist can help when a few meaningful alternatives need comparison. An explicit archive becomes interesting when the surrounding conditions change and previously useful forms deserve to remain available.

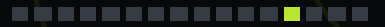
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Choose the output the decision needs

For the learning example, a person might begin with a worked exercise and later want an open investigation. That change does not prove the exercise was a mistake. It gives another region of the repertoire a job. The product then needs a way to find and revisit alternatives. Generating them cheaply is only part of the work. Preserve why each candidate might be useful, what it assumes and which conditions would make it worth considering again.

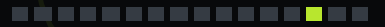
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Investigate an empty region

An empty area of the map is an invitation to think, not automatic evidence of an opportunity. Perhaps the team has not explored it. Perhaps the axes describe a combination that makes little sense. Perhaps producing that kind of candidate is unusually difficult.

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Investigate an empty region

Try drafting a concrete candidate for the gap. If you cannot say what a person would do with it, reconsider the map before investing in a more elaborate search. A technically diverse collection can still be irrelevant to the actual decision.

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Investigate an empty region

The archive also has a cost. Candidates can become stale, comparisons can become burdensome and the chosen distinctions can lose relevance. Retain an alternative because it serves a plausible purpose, not because deleting anything feels intellectually narrow.

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Investigate an empty region

For your next design discussion, bring both a ranking and a map of the same candidates. Ask which useful difference the ranking conceals. The answer may tell you that one winner is enough—or reveal a product that needs a repertoire.